

# Design Properties and Code Quality

Discussion 09 – CSC 4033

## Readings

The reading selections are from two books: “[Software Engineering: A Modern Approach](#)” (web version available for free online), by Marco Valente, and “Refactoring: Improving The Design of Existing Code” (viewable on Internet Archive), by Martin Fowler. Both reading selections run the risk of simply being lists of facts, so do your best to tie in relevant experiences to the discussion!

## Required

- 1) [Software Engineering: A Modern Approach](#), Ch.5, Intro-5.5 (~17 pages)
- 2) Refactoring: Improving The Design of Existing Code, Ch.3 (8 pages)

## Recommended

- 1) [Design Principles in System Design - GeeksforGeeks](#)
- 2) Even though real examples are given in reading selections, I recommend supplementing those with others you’ve found out in the wild to enrich the discussion.

## Learning Outcomes

- Discuss at least five design properties and give a real example of each
- Identify a project you’ve worked on before that either *did* incorporate a specific design property or *could have* benefited from incorporating one. Explain
- Explain what code smells are for (and what they aren’t for) and give at least five examples
- Identify a project you’ve worked on before (or one you have seen) whose code had “bad smells”. What smells were present and how significantly did they diminish the code quality?